

AV09: Data Visualisation & Open Source Technology Advance Session: {teal} as a Catalyst for Cross-Industry Collaboration and Innovation in Clinical Insight Generation

Phuse Connect US 2026

Dony Unardi
Vedha Viyash



It's all Shiny Modules!

What are the technical skills of teal developers?

Teal app developer

- Ability to install and use R packages
- Data wrangling with R
- Basic knowledge of Shiny (optional)
- Ability to work with ADaM datasets
- Understand teal_data/teal_data_module

Teal framework developer

- R package development skills
- General software development skills
- Deeper understanding of the teal framework

Teal module developer

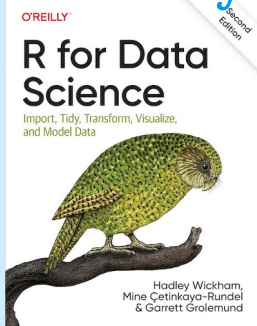
- R package development skills
- Data wrangling with R
- Data visualization with R
- Advanced knowledge of Shiny
- Deep understanding of teal_data

Learning path for Teal app developer

General skills



r4ds.hadley.nz



Teal specific skills

Data input

insightsengineering.github.io/teal

A screenshot of the teal GitHub repository page. A red box highlights the 'Articles' dropdown menu, which lists various articles including 'Getting Started with teal', 'Data in 'teal' apps', 'Including Data in teal Applications', 'Data as shiny Module', 'Filter Panel', 'Transform Input Data', 'Extending 'teal'', 'Creating Custom Modules', 'Adding Support for Reporting to Custom Modules', 'Transform Module Output', 'Using 'teal'', 'Teal as a Shiny Module', 'Modifying a teal Application With R Options', and 'Bootstrap Themes in teal'. Red arrows point from the text 'Very important' to the 'Articles' dropdown and the 'Including Data in teal Applications' article.

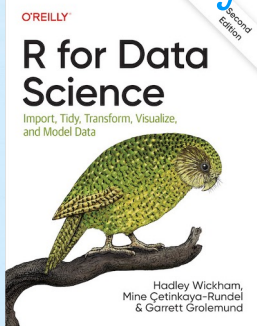
Very important

Learning path for Teal app developer

General skills



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Teal specific skills

Data input

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teal 0.16.0 part of Get started Reference Articles Technical blueprint Changelog Reports Versions Search for

teal: Interactive Exploration Analysis with Shiny Web Applications

CRAN 0.16.0 downloads 8763 downloads 501/month downloads

Check passing Docs passing Test Coverage 56-50%

Forks 45 Stars 210

commit activity 4/month contributors 36 last commit 1 day pull requests

repo status Active package version v0.16.0.9006 open issues 65

teal is a shiny-based interactive exploration framework that requires app developers to specify:

- Data, which can be:
 - CDISC data, commonly used for clinical trial reporting
 - Independent datasets, for example from a data frame
 - Related datasets, for example a set of data frames
 - MultiAssayExperiment objects which are R data structures for representing and analyzing multi-omics experiments
- teal modules:
 - teal modules are shiny modules built within the teal framework that specify analysis to be performed. For example, it can be a module for exploring outliers in the data, or a module for visualizing the data in line plots. Although these can be created from scratch, many teal modules have been released and we recommend starting with modules found in the following packages:
 - teal.modules.general: general modules for exploring relational/independent/CDISC data
 - teal.modules.clinical: modules specific to CDISC data and clinical trial reporting
 - teal.modules.hermes: modules for analyzing MultiAssayExperiment objects

Get started

- Getting Started with teal
- Data in 'teal' apps
 - Including Data in teal Applications
 - Data as shiny Module
- Filter Panel
- Transform Input Data
- Extending 'teal'
 - Creating Custom Modules
 - Adding Support for Reporting to Custom Modules
 - Transform Module Output
- Using 'teal'
 - Teal as a Shiny Module
 - Modifying a teal Application With R Options
 - Bootstrap Themes in teal
- More articles...

Links

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)

License

Apache License 2.0

Community

- [Contributing guide](#)
- [Code of conduct](#)

Citation

- [Citing teal](#)

Developers

- Dawid Kaledkowski
 - Author, maintainer
- Pawel Rucki
 - Author
- Aleksander Chlebowski
 - Author
- Andre Verissimo
 - Author
- Kartikya Kiran
 - Author
- Vedha Vijayash
 - Author
- Marcin Kosinski
 - Author

Very important

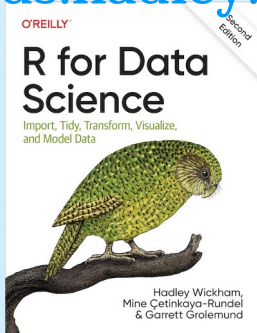
Next step

Learning path for Teal app developer

General skills



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Teal specific skills

Using modules

insightsengineering.github.io/teal.modules.general

17 modules

General data

insightsengineering.github.io/teal.modules.clinical

37 modules

CDISC ADaM data

insightsengineering.github.io/teal.modules.hermes

8 modules

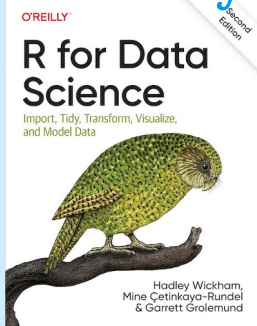
MultiAssayExperiment data

Learning path for Teal app developer

General skills



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Teal specific skills

Using modules

A screenshot of the teal.modules.clinical documentation page. The page has a navigation bar with links: Get started, Reference (highlighted with a red box), Articles, Changelog, Reports, and Versions. A search bar is on the right. The main content lists various functions under the heading 'Graphs'. A red arrow points from the 'Reference' tab to the 'tm_g_km()' function, which is also highlighted with a red box. The function is described as 'teal Module: Kaplan-Meier Plot'. Other functions listed include tm_t_summary_by(), tm_t_tte(), tm_g_barchart_simple(), tm_g_ci(), tm_g_forest_rsp(), tm_g_forest_tte(), tm_g_ipp(), tm_g_lineplot(), tm_g_pp_adverse_events(), tm_g_pp_patient_timeline(), tm_g_pp_therapy(), and tm_g_pp_vitals().

teal.modules.clinical 0.10.0 part of

Get started **Reference** Articles Changelog Reports Versions

Search for

tm_t_summary_by()
teal Module: Summarize Variables by Row Groups

tm_t_tte()
teal Module: Time-To-Event Table

Graphs

tm_g_barchart_simple()
teal Module: Simple Bar Chart and Table of Counts per Category

tm_g_ci()
teal Module: Confidence Interval Plot

tm_g_forest_rsp()
teal Module: Forest Response Plot

tm_g_forest_tte()
teal Module: Forest Survival Plot

tm_g_ipp()
teal Module: Individual Patient Plots

tm_g_km()
teal Module: Kaplan-Meier Plot

tm_g_lineplot()
teal Module: Line Plot

tm_g_pp_adverse_events()
teal Module: Patient Profile Adverse Events Table and Plot

tm_g_pp_patient_timeline()
teal Module: Patient Profile Timeline Plot

tm_g_pp_therapy()
teal Module: Patient Profile Therapy Table and Plot

tm_g_pp_vitals()

On this page

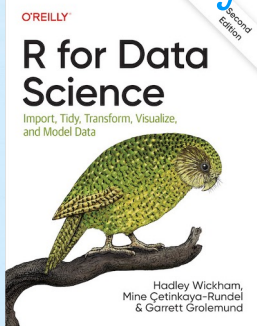
- Tables
- Graphs
- General analyses
- Patient profile modules
- Helper functions
- Data

Learning path for Teal app developer

General skills



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Teal specific skills

Using modules

teal.modules.clinical 0.10.0 part of teal.modules

Get started Reference Articles Changelog Reports Versions Search for

Examples in Shinylive

example-1 [Open in Shinylive](#)

On this page Usage

Shiny for R

Kaplan-Meier Plot

Report preview

Reporter

Encodings

Datasets: ADSL, ADTTE

Select Endpoint

Dataset: ADTTE

Filter by

OS Overall Survival

Analysis Variable

Select AVAIL

Censor Variable

Select CENS

Facet Plots by

Dataset: ADSL

Select Treatment Variable

Dataset: ADSL

Select

AFM Description of Plan

KM Plot of OS Stratified by SEX

Survival Probability

Time (Days)

1695% Relative Survival

A: Drug X (0.647, 1.06) 0.0529

C: Combination 1 (0.886, 1.92) 0.1881

Patients at Risk:

	0	1000	2000	3000	4000
B: Placebo	73	18	3	1	1
A: Drug X	69	22	4	2	1
C: Combination	58	11	1	0	0

Test for Graphical Proportion and p-value Method for Logrank-Ratio Test

Active Filter Summary

Data Name	Obs	Subjects
ADSL	200/200	200/200
ADTTE	1000/1000	200/200

Active Filter Variables

ADSL +

ADTTE +

Examples

```
library(nestcolor)

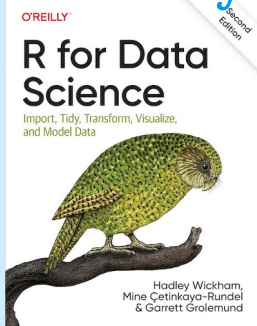
data <- teal_data()
data <- within(data, {
```


Learning path for Teal app developer

General skills



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Teal specific skills

Transform data

insightsengineering.github.io/teal

teal 0.16.0 part of Get started Reference

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Very important

Next step

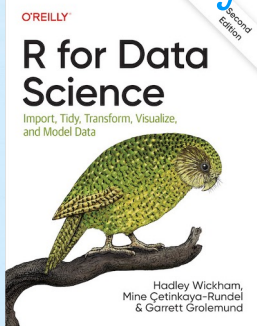
Final step

Learning path for Teal app developer

General skills



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Teal specific skills

Decorate output

insightsengineering.github.io/teal.modules.clinical

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Get started Reference **Articles** Changelog Reports Versions Search for

Decorate Module Output

NEST CoreDev

Source: [vignettes/decorate-module-output.Rmd](#)

Introduction

The outputs produced by teal modules, like graphs or tables, are created by the module developer and look a certain way. It is hard to design an output that will satisfy every possible user, so the form of the output should be considered a default value that can be customized. Here we describe the concept of *decoration*, enabling the app developer to tailor outputs to their specific requirements without rewriting the original module code.

The decoration process is build upon transformation procedures, introduced in teal. While transformers are meant to edit module's input, decorators are meant to adjust the module's output. To distinguish the difference, modules in teal.modules.clinical have 2 separate parameters: transformers and decorators.

To get a complete understanding refer the following vignettes:

- Transforming the input data in [this vignette](#).
- Transforming module output in [this vignette](#).

Outputs that can be decorated

It is important to note which output objects from a given module can be decorated. The module function documentation's *Decorating Module* section has this information.

You can also refer the table shown below to know which module outputs can be decorated.

On this page

- Introduction
- Outputs that can be decorated
- Decorating listing_df, ElementaryTable, TableTree
- Decorating ggplot
- Decorating datatables

Articles

- Decorate Module Output
- Example Data Generation
- Quick start: 'substitute' for NSE

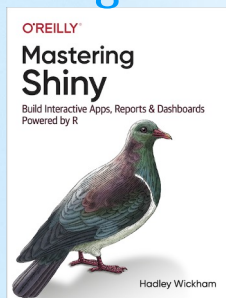
One step
further

Learning path for Teal module developer

General skills

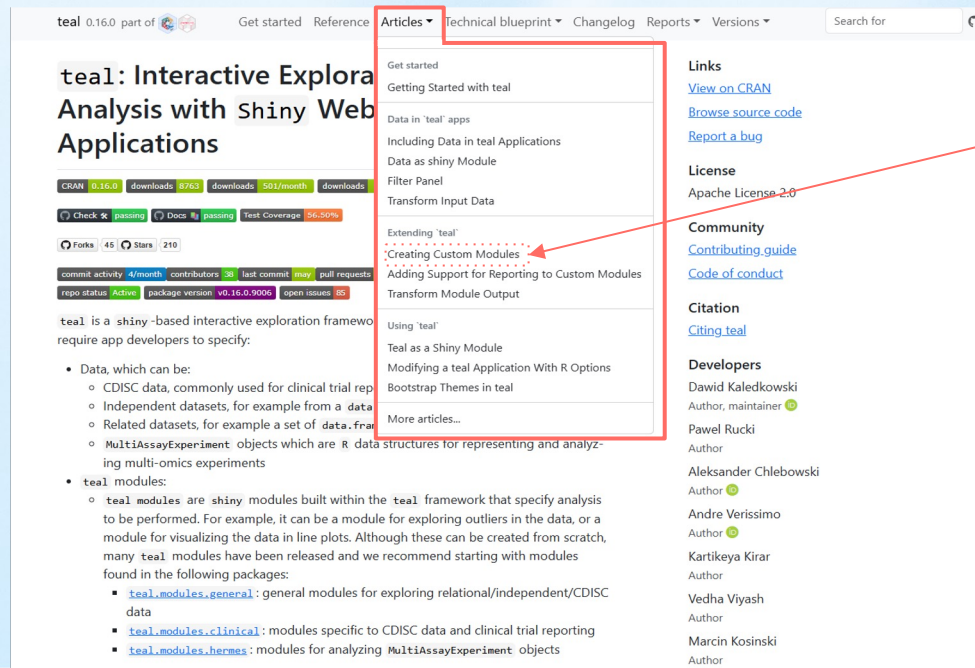


mastering-shiny.org



Teal specific skills

insightengineering.github.io/teal



Create module

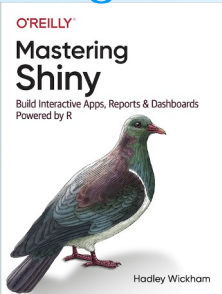
Very important

Learning path for Teal module developer

General skills

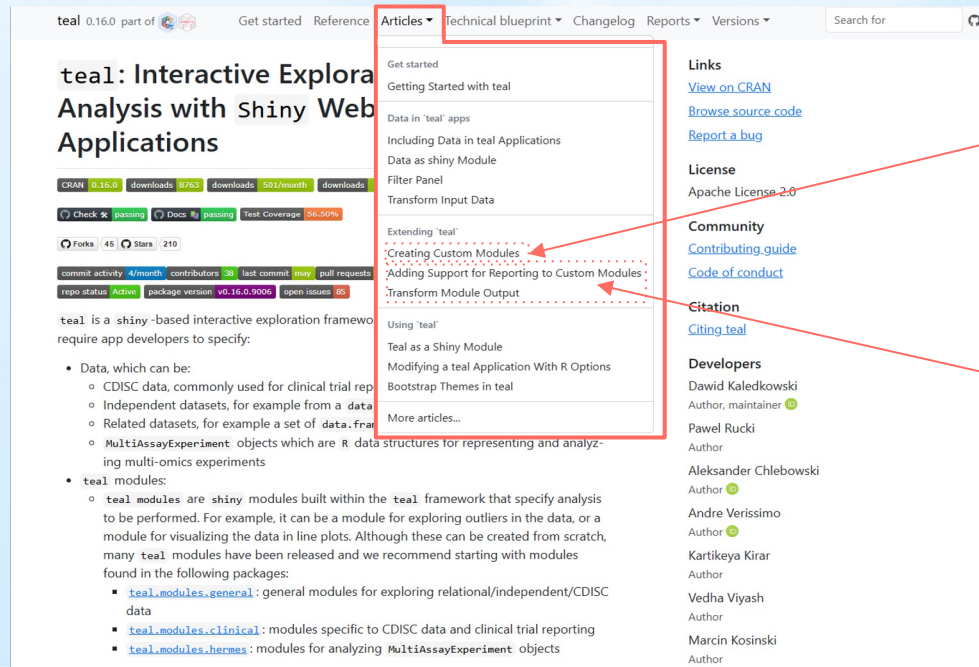


mastering-shiny.org



Teal specific skills

insightsengineering.github.io/teal



Add features

Very important

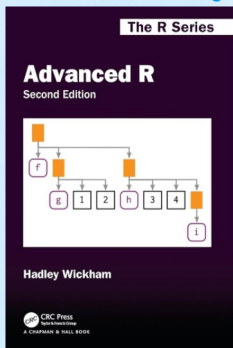
Extending module

Learning path for **Teal framework developer**

General skills



adv-r.hadley.nz



Teal specific skills

Understand and use/extend core objects of teal framework

- teal: [ui_teal](#), [srv_teal](#), [teal_data_module](#), [teal_transform_module](#)
- teal.data: [teal_data](#)
- teal.slice: [FilterPanelAPI](#)

Should have gone through all the teal vignettes to understand how the framework works.